

A Hand-based Collaboration Framework in Egocentric Coexistence Reality

Jeongmin Yu¹, Seungtak Noh², Youngkyoon Jang³, Gabyong Park⁴ and Woontack Woo⁵

^{1,2} KAIST CTRI, ^{3,4,5} KAIST UVR Lab., Daejeon, Korea

(Tel : +82-42-350-5923; E-mail: jmyu119, stnoh, y.jang, gypark, wwoo@kaist.ac.kr)

Abstract - This paper presents a hand-based collaboration framework in egocentric coexistence reality (CR), which enables a local user to collaborate with multiple users in the distance. To develop the remote collaboration system, we integrate the following core technologies: i) hand-augmented object interaction method in CR environment by tracking the bare hands or hands with grasping real-world objects in egocentric view, ii) a remote user's hand-body information fusion method for enhancing spatial co-presence in CR environment. According to preliminary experimental results, we confirm the feasibility as a future remote collaboration platform in egocentric CR.

Keywords – Hand-object interaction, remote collaboration, coexist reality, hand and body information fusion.

1. Introduction

Seamless collaboration with the remote persons is one of the most significant challenges in computer-supported cooperative working environment. To give a immersive sense of co-operations, an efficient collaboration system which enables a local person to mutually share, interact, and collaborate together with remote persons is needed.

For the past few decades, several approaches for seamless remote collaboration using networked system have been proposed such as video teleconference system, screen-based system, and virtual avatar-mediated system. The video teleconference systems are bringing remote audiences to communicate and interact as if they are in the same physical site [3, 4]. These systems make a shared common virtual environment so that the virtual audiences can follow the presentation and interact with a real presenter. However, these systems are just interact with other remote audiences for learning and discussion, they do not actually cooperate a shared target task. In addition, touch screens or digital whiteboards have been used to support collaboration between remote persons. Using these systems, several researchers attempt to enhance co-located collaboration with distributed remote persons [5, 6], but they only perform the limit interactions in the spatially confined 2D/3D displays. Meanwhile, virtual avatar as a remote user is another representative method for displaying remote user. The most of remote collaboration researches using the avatar are performed on a shared virtual workspace [7, 8].

In this paper, we propose a novel hand-based collaboration framework in egocentric coexistence reality

(CR), which enable a local user to collaborate together with remote users. The framework consists of two main partial system: i) a robust real-time bare hands tracking and gesture recognition system to partial occlusions due to own hands or fingers and real-world objects, ii) hand-body information fusion system which combines hand and full-body information tracked by an egocentric movable camera on head-mounted display (HMD) and an exocentric fixed RGB-D camera, respectively.

By adopting the hand tracking and gesture recognition methods in egocentric viewpoint, the proposed system supports elaborate interaction between hand and shared 3D virtual object. Furthermore, our system provides the sense of spatial coexistence to the local user who works with remote users owing to realistic avatar's movement. The proposed system has no dependency on specific application domain, it can be possible to apply in many HMD-based remote collaborations.

2. Scenario of Remote Collaboration in CR

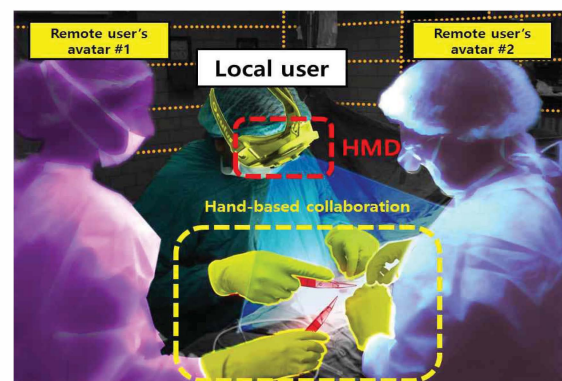


Fig. 1. Illustration of hand-based collaboration in egocentric coexistence reality.

In egocentric CR space, a local user can mutually share, interact, collaborate, and feel the co-presence with physically remote users under connected real world, virtual world and remote world. The proposed hand-based collaboration system in egocentric CR can be applied to performing on remote co-surgery or education, as shown in Fig. 1. One surgeon in a local site is wearing a see-through HMD while other two co-surgeons who reside in remote sites are using same devices. The local surgeon in real world site is able to see other two remote co-surgeons as the virtual avatars and shared virtual 3D objects. They can collaborate the shared target task by bare hands or hands with grasping surgical instruments.

Although the surgery takes place only one physical workspace, the remote co-surgeons can manipulate augmented virtual objects together and feel the sense of co-presence.

3. Proposed Framework

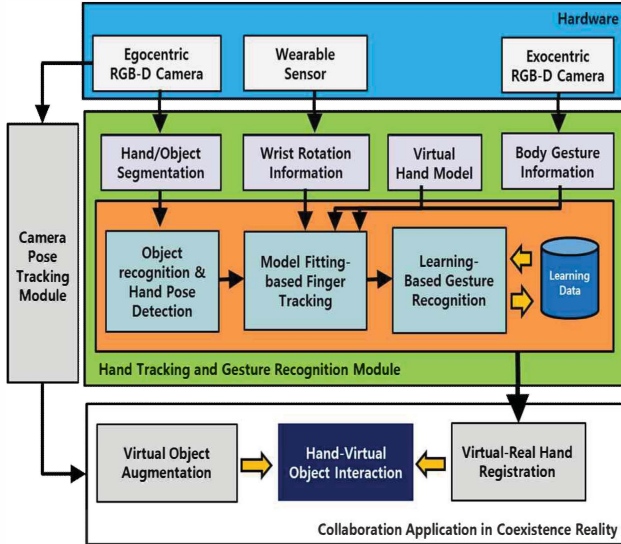


Fig. 2. System diagram of hand-based collaboration in egocentric coexistence reality

Figure 2 shows the overall system diagram of hand-based collaboration in egocentric CR environment. For this system, we use an egocentric short-range RGB-D camera and wearable sensors for accurate hands (fingers) tracking and gesture recognition, and exocentric RGB-D camera for full-body tracking. From the egocentric RGB-D images, we first segment bare hands and grasping real-world objects, which are used to object recognition and hand pose detection. Then, hands and fingers are tracking based on a model fitting method. Finally, hand gesture recognition using the tracked hand information is performed by a learning-based method. After registration between virtual and real hands, we can interact with a 3D virtual object for a shared target task. The detail methodological descriptions of hand-augmented object interaction and hand-body information fusion are presented as follows.

3.1 Hand-augmented Object Interaction

The tracking of hand articulations is essential for manipulating an augmented object. For this, we use a hybrid algorithm adopting detection and tracking methods. In detection part, parts as each finger and palm are classified from trained model in offline procedure under challenging situations like severe occlusions. This classification procedure reduces the ratio of wrong correspondence between the predefine hand model and observation from camera. Then, the tracking algorithm is conducted after classifying each part of the hand. The tracking algorithm makes the solution fall to less local optimal solution than conventional tracking algorithms,

because the observation from camera sensor has 3D geometric information and specific label.

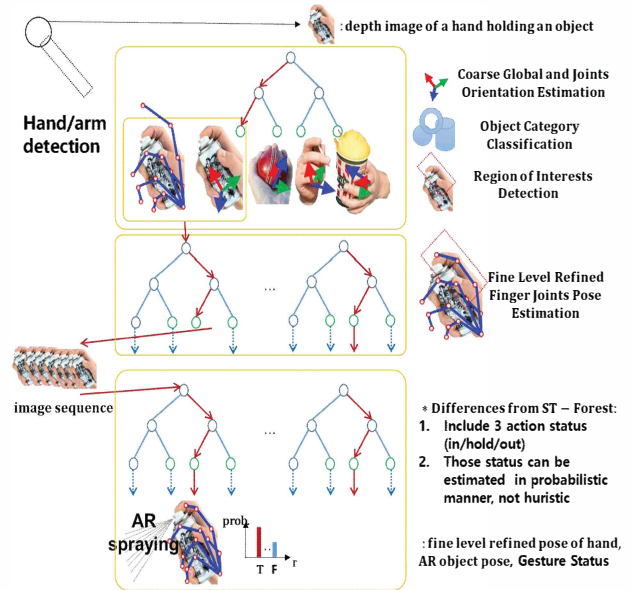


Fig. 3. Multi-layered kinematic constrained spatio-temporal (KCST) forest.

In addition to the accurate hand posture recognition, we propose a real-time bare hand gesture estimation, which promises several possibilities for augmented reality (AR) and virtual reality (VR) interaction scenarios, such as gesture input in a wearable AR [1] and direct object-selection interaction [2]. In the computer vision domain, although there have been studies about hand posture recognition in the process of hand-object interactions, the most recent methods [9] and [10] are insufficient to be applied directly to this work, due to the following challenges of the task: occlusions in hand-object interaction, variances of manipulating motion, ambiguousness of hand and object regions, and unnatural gesture mapping with bare hand.

Addressing these challenges, a novel Kinematics Constraint-based Spatio-Temporal (KCST) forest for 3D hand posture estimation and specified finger gesture status recognition is proposed, as shown in Fig. 3. The KCST forest learns from the sequence of the accurate position of hand joints, which are detected by estimating a global hand posture and classifying an object category. Even though both estimated global hand posture and classified object category are somewhat noisy due to ambiguity of hand and regions, by specifying the finger region based on the kinematic constant, the proposed KCST forest is able to finely refine the joint positions of interesting. This work aims to infer an AR object corresponding to a real object as well as its manipulating action status, based on hand posture holding an interactive object and its joint movements. The expected contributions of this work are threefold: invariant to the rotation and translation of hand, kinematics constrained spatio-temporal information learning, robust bare hand-based AR object manipulation holding a real abstract object.

3.2 Hand-body Information Fusion

In this work, we support a collaboration between local and remote user. Unlike an interaction system for single user, we need to consider how to share a collaborative space and object, and how to achieve collaboration with high level co-presence. To achieve these considerations, we utilize a virtual avatar as a surrogate of remote user and summon it in local user's space.

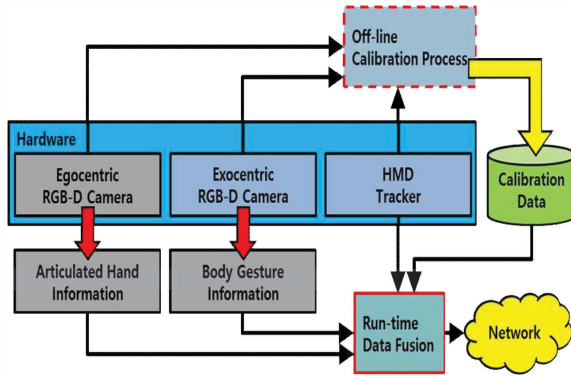


Fig. 4. Schematic diagram of hand-body fusion

For implementing realistic avatar movement, we need not only head and hand but also full-body movement information. In typical HMD-based hand interaction system without exocentric camera, the available movement is floating head and hands on the mid-air like ghost, because of the lack of body tracking information. Due to its unrealistic impression, it is a reason of low co-presence in collaboration. Thus, we adopt exocentric RGB-D camera for full-body tracking. Figure 4 shows the schematic diagram of our approach.

The main issue on this idea is that the source of hand and body tracking are totally different: hand tracking is done by egocentric data, while body tracking is done by exocentric RGB-D camera. In addition, HMD is always moving by following user's movement, so we cannot assume that those two cameras have a fixed relationship. It can also be possible to think to track the hand in exocentric view, but it is not promising approach because of low resolution.

To acquire dynamic relationship, we also adopt HMD tracker and utilize it as intermediate transformation to obtain egocentric camera pose in exocentric camera coordinate system. By means of HMD tracker, getting dynamic camera pose problem becomes getting two fixed relative poses: exocentric camera to HMD tracker and HMD to egocentric camera. For obtaining these poses, we need off-line calibration stage before hand-body fusion. If we uses exocentric camera as the HMD tracker also, the former might not be considered.

Once calibration is done, we can execute hand-body information fusion process. Even though we can acquire dynamic relationship between egocentric and exocentric camera, there is a still open problem how to fuse the body and hand tracking data together. The fusion method should consider the tracking error and tracking lost by hardware limitation such as range and field-of-view (FOV).

4. Experimental Settings and Results

4.1 Experimental Setting

We adopt the off-the-shelf hardware for building our experimental system. We use an Oculus DK2 HMD with an OvrVision stereo camera for video see-through. Additionally, we attached Creative Senz3D as an ego centric RGB-D camera for hand tracking. For hand tracking, we used the Intel Hand Skeletal Library which is based on Melax et al. [11]. With this tracker, we also attached clicking detection and position estimation module [2]. We also set a Microsoft Kinect2 in front of the user for body tracking. Overall software is integrated in the Unity Engine, and local and remote side are linked with local network.

The system consists of most mandatory modules, but some part are still not implemented yet. The calibration between exocentric camera and HMD is done by ad-hoc approach with user-defined value.

4.2 Preliminary Results

The experiment for the hand tracking method is conducted firstly on a synthetic dataset which is made from rendered depth images generated by the hand's pose in OpenGL. We calculate an error between a ground-truth and an estimated fingertip position of each finger, as shown in Fig. 5. From the experiment, we observed that the tracking error of the thumb (i.e., 16mm) was bigger than other fingers. This is because the metacarpophalangeal (MCP) joint of the thumb is included in a palm's region unlike other joints.

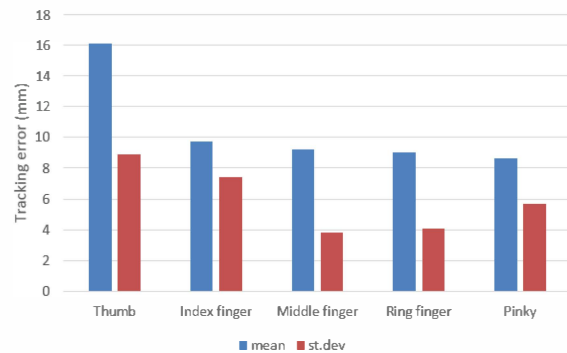


Fig. 5. Results of hand tracking in synthetic dataset.

The second experiment for the hand gesture recognition method is tested on a gesture dataset [2] which contains the clicking actions with each finger. From the experiment, we observed that the clicking action detection accuracies of thumb, index finger, middle finger, ring finger, and pinky show 89.80%, 96.90%, 95.68%, 92.82% and 94.37%, respectively. The reason why the accuracy of clicking action of thumb is lower than other fingers is due to the tracking failure of the hand skeletal model.

Lastly, figure 6 shows a snapshot of user's perspective from our preliminary implementation. In this figure, local and remote users collaborate on blue and yellow colored virtual blocks. The remote user is visible in local user's FOV to confirm the consistency with avatar motion. Even

though our system is not fully implemented, we can validate our approach is feasible and convincing enough for remote collaboration.

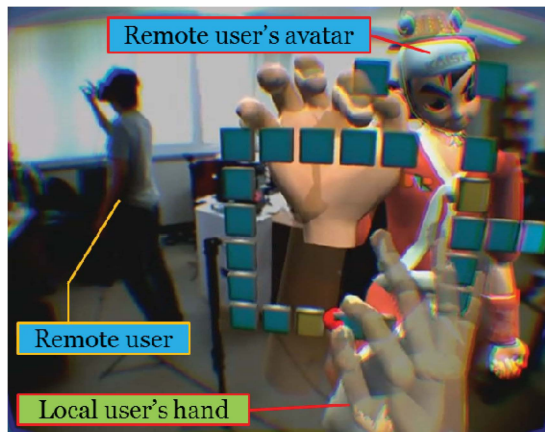


Fig. 6. A snapshot from local user's view. Remote user (left) is summoned as a virtual avatar (right) in front of local user. Local user can manipulate shared virtual object (center) by one's hand (bottom).

5. Conclusion and Future Work

In this paper we have presented a hand-based remote collaboration system in egocentric CR, which allows a local user to collaborate a shared task with other users who worked in physically remote sites. Utilizing our system, the local user wearing a HMD can manipulate a shared virtual 3D objects with remote users by bare hand. In addition, the virtual avatar as the remote user which represented by hand-body information fusion method enables the local user to feel the sense of coexistence.

Preliminary results show its strong possibility as a future remote collaboration platform. As the future works, we will focus on the development of robust real-time hand tracker under severe occlusions and hand-body fusion method using an automatic calibration technique.

Acknowledgement

This work was supported by the Global Frontier R&D Program on <Human-centered Interaction for Coexistence> funded by the National Research Foundation of Korea grant funded by the Korean Government(MSIP) (2010-0029751) and Ministry of Culture, Sports and Tourism(MCST) and Korea Creative Content Agency(KOCCA) in the Culture Technology(CT) Research & Development Program 2014.

References

- [1] A. Colac, A. Kirmani, H. S. Yang, N.-W. Gong, C. Schmandt and V. K. Goyal, "Mime: Compact, low power 3D gesture sensing for interaction with head mounted displays," In proc. of UIST, pp. 227-236, 2013.
- [2] Y. Jang, S.-T. Noh, H. J. Chang, T.-K. Kim and W. Woo, "3D Finger CAPE: Clicking Action and Position Estimation under Self-Occlusions in Egocentric

- Viewpoint," IEEE Transactions on Visualization and Computer Graphics, Vol. 21, No. 4, pp. 501-510, 2015.
- [3] C. Kuster, T. Popa, J.-C. Bazin, C. Gotsman and M. Gross, "Gaze Correction for Home Video Conferencing," ACM Transactions on Graphics, Vol. 31, No. 6, 2012.
- [4] G. Sharma and Ralph Schroeder, "Mixing real and virtual conferencing," Virtual Reality, Vol. 17, No. 3, pp 193-204, 2013.
- [5] S. Beck, A. Kunert, A. Kulik and B. Froehlich, "Immersive Group-to-Group Telepresence," IEEE Transactions on Visualization and Computer Graphics, Vol. 19, No. 4, pp. 616-625, 2013.
- [6] K. Higuchi, Y. Chen, P. A. Chou, Z. Zhang and Z. Liu, "ImmerseBoard: Immersive Telepresence Experience using a Digital Whiteboard," In proc. of CHI, pp. 2383-2392, 2015.
- [7] R. Schroedera, A. Steedb, A. Axelssona, I. Heldala, A. Abelinc, J. Wideströmd, A. Nilssona and M. Slaterb, "Collaborating in networked immersive spaces: as good as being there together?," Computers & Graphics, Vol. 23, No. 5, pp. 781-788, 2001.
- [8] W. Steptoe, J. Normand, O. Oyekoya, F. Pece, E. Giannopoulos, F. Tecchia, A. Steed, T. Weyrich, J. Kautz and M. Slater, "Acting Rehearsal in Collaborative Multimodal Mixed Reality Environments," Presence, Vol. 21, No. 4, pp 406-422, 2012.
- [9] N. Kyriazis and A. Argyros, "Scalable 3D tracking of multiple interacting objects," In proc. of CVPR, pp. 3430-3437, 2014.
- [10] G. Rogez, M. Khademi, J. Supancic, J. Montiel and D. Ramanan, "3D hand pose detection in egocentric RGB-D images," In proc. of ECCV Workshop, pp. 356-371, 2014.
- [11] S. Melax, L. Keselman and S. Orsten, "Dynamics based 3D skeletal hand tracking," In proc. of Graphics Interface, pp. 60-70, 2013.